

Exploratory Data Analysis Report

HAM10000 Skin Lesion Dataset

Multi-class Skin Lesion Classification — Dermoscopic Image Dataset

1. Dataset Overview

The HAM10000 ("Human Against Machine with 10000 training images") dataset is a collection of dermoscopic images used for multi-class skin lesion classification. This report summarizes the exploratory data analysis (EDA) performed on the dataset's metadata and image properties, covering class balance, patient demographics, image quality, and pixel-level color statistics.

1.1 Raw Metadata Sample

The metadata table has 10,015 rows and 8 columns (lesion_id, image_id, dx, dx_type, age, sex, localization, plus the derived image_path). A sample of the first and last rows:

lesion_id	image_id	dx	dx_type	age	sex	localization
HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp
HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp
HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp
...	...	...	...	...	...	...
HAM_0000239	ISIC_0032854	akiec	histo	80.0	male	face
HAM_0003521	ISIC_0032258	mel	histo	70.0	female	back

1.2 Dataset Summary

Property	Value
Dataset Name	HAM10000
Problem Type	Multi-class Skin Lesion Classification
Total Images	10,015
Number of Classes	7
Image Format	JPG
Metadata Columns	8
Missing Values	57 (all in 'age')
Duplicate Rows	0

2. Missing Values

Only the age column contains missing entries — 57 out of 10,015 records (0.57%). All other metadata fields (lesion_id, image_id, dx, dx_type, sex, localization) are fully populated.

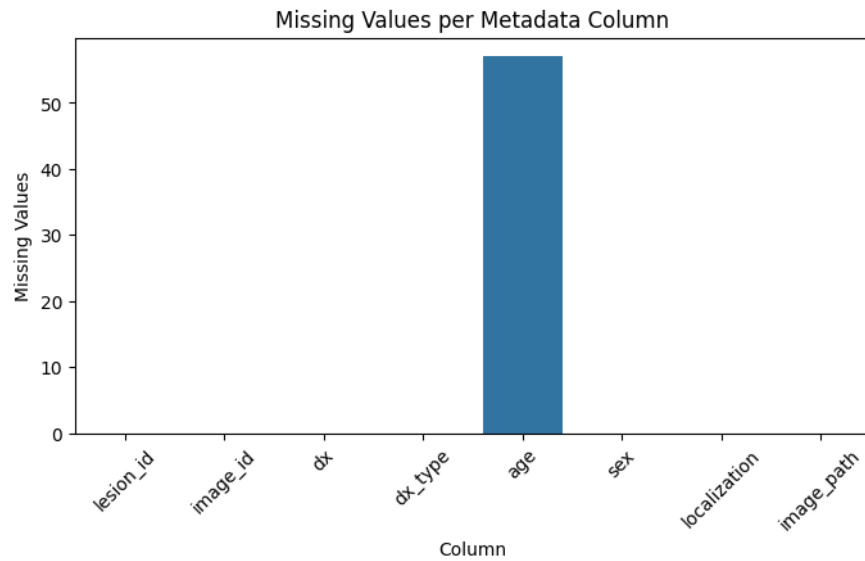


Figure 1. Missing values per metadata column

3. Patient Demographics

3.1 Age Distribution

Patient age is roughly centered in the 30–60 range, with a peak around 40–45 years old, and relatively few pediatric cases.

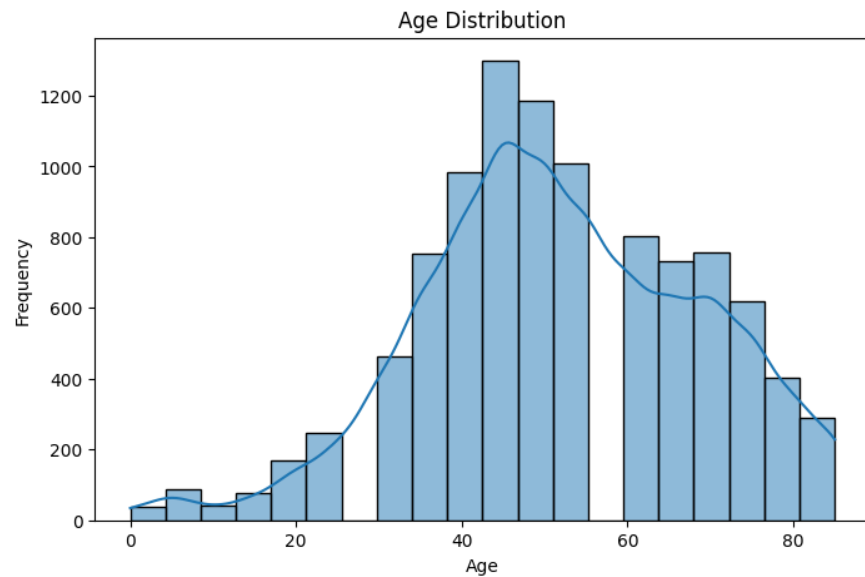


Figure 2. Age distribution of patients

3.2 Sex Distribution

The dataset skews male, with more male-labelled records than female.

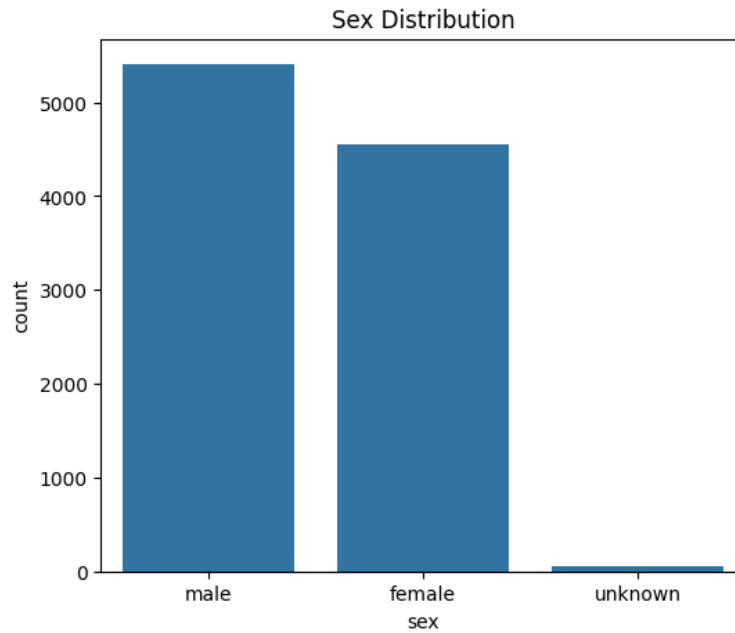


Figure 3. Sex distribution

3.3 Lesion Localization

Lesions are most frequently recorded on the back, lower extremity, trunk, and upper extremity, with the face and scalp also well represented; sites such as acral and genital regions are rare.

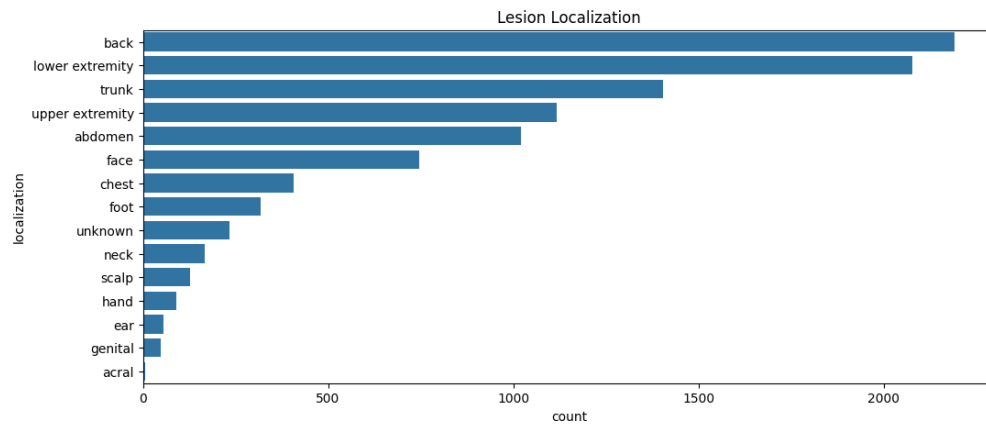


Figure 4. Lesion localization across body sites

4. Duplicate Lesions

The dataset contains 2,545 duplicated lesion_id entries — this is expected, since HAM10000 includes multiple dermoscopic images per unique lesion (captured at different angles/times). The most repeated lesions appear up to 6 times.

Lesion ID	Image Count
HAM_0003789	6
HAM_0005263	6

Lesion ID	Image Count
HAM_0000835	6
HAM_0001863	6
HAM_0000893	5

This repetition means a simple random train/test split risks leaking images of the same lesion across splits; grouping by lesion_id is recommended when partitioning data.

5. Class Balance Analysis

5.1 Class Counts & Distribution

Class	Full Name	Count	Percentage
nv	Melanocytic Nevus	6,705	66.95%
mel	Melanoma	1,113	11.11%
bkl	Benign Keratosis	1,099	10.97%
bcc	Basal Cell Carcinoma	514	5.13%
akiec	Actinic Keratoses	327	3.27%
vasc	Vascular Lesion	142	1.42%
df	Dermatofibroma	115	1.15%

Majority class: nv (6,705 images) **Minority class:** df (115 images) **Imbalance ratio:** 58.30 : 1

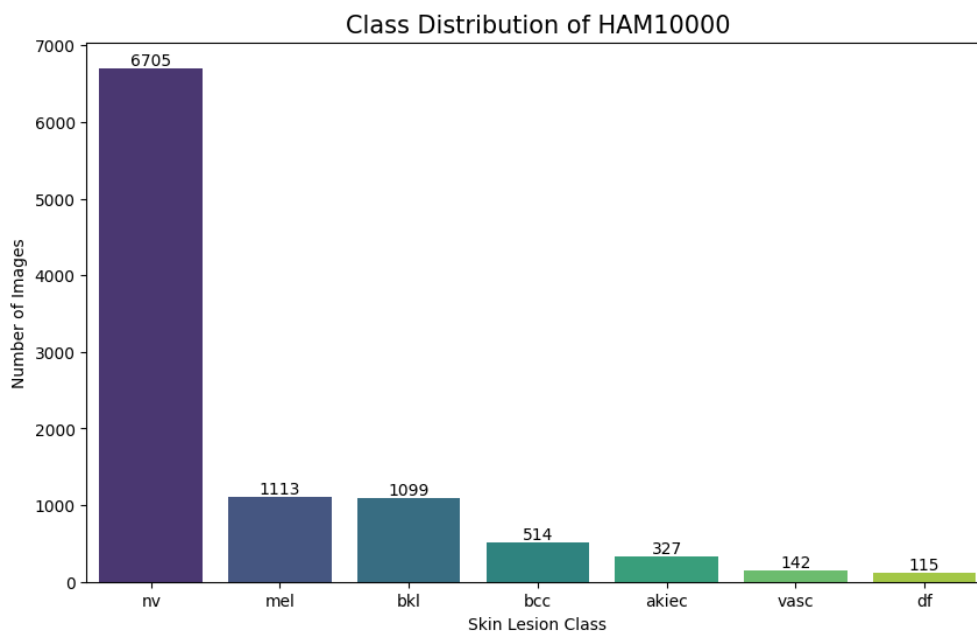


Figure 5. Class distribution (image count per class)

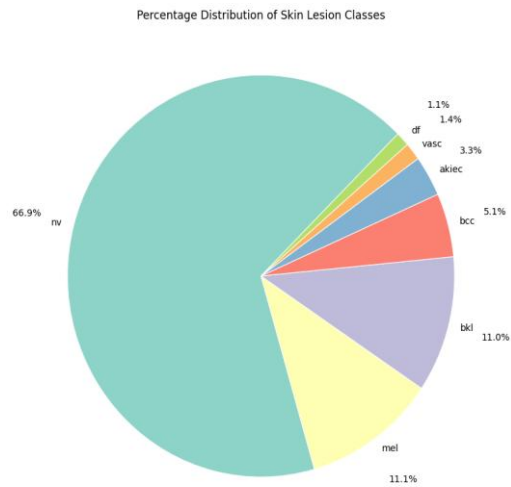


Figure 6. Percentage distribution of classes

6. Labelled Sample Grid

Five representative sample images were drawn from each of the seven classes to visually inspect inter-class similarity and intra-class variability.

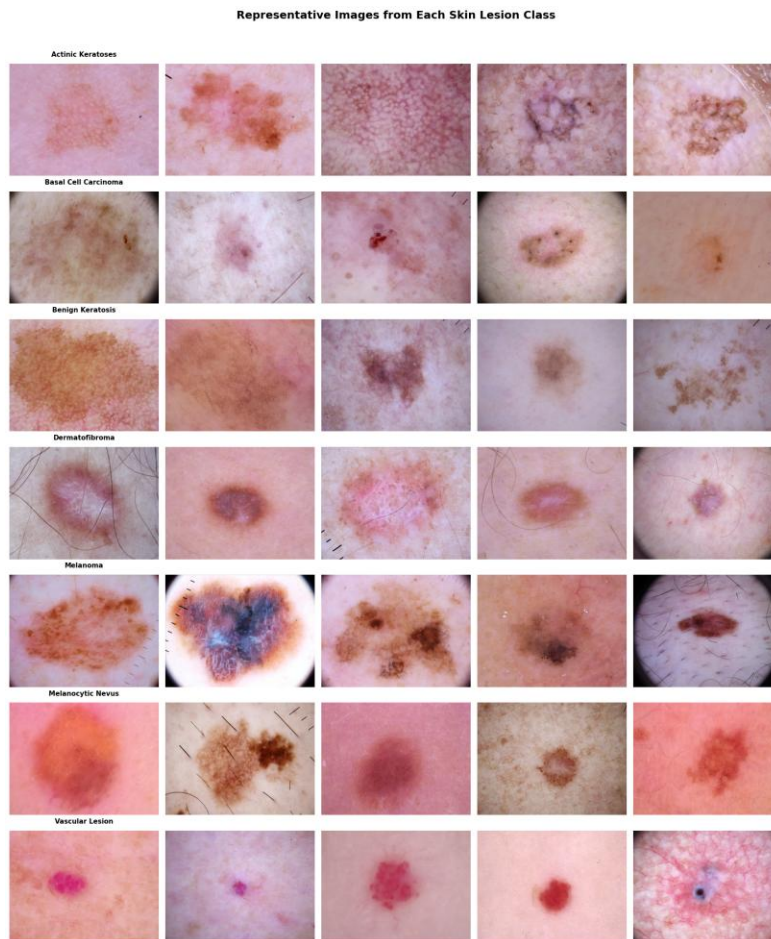


Figure 7. Sample images per lesion class (rows = classes, columns = samples)

7. Image Size & Resolution Analysis

Every one of the 10,015 images shares an identical resolution of 600×450 pixels, with zero variance across the dataset. This uniformity removes the need for resizing-related preprocessing decisions (aside from any deliberate downscaling for model input).

Statistic	Width	Height
count	10,015	10,015
mean	600.0	450.0
std	0.0	0.0
min	600.0	450.0
25%	600.0	450.0
50%	600.0	450.0
75%	600.0	450.0
max	600.0	450.0

A groupby on (width, height) confirms a single unique resolution: 600×450 , covering all 10,015 images.

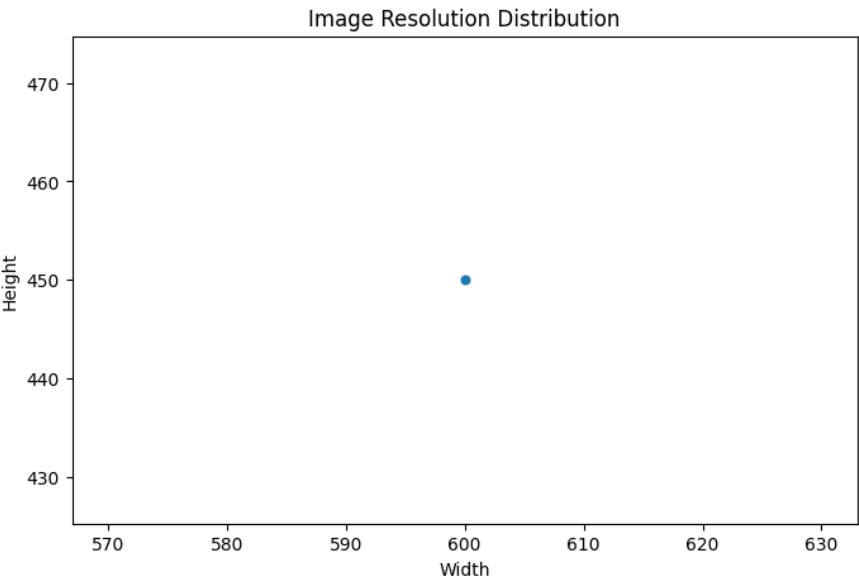


Figure 8. Width vs. height scatter — confirms a single fixed resolution

8. Pixel & Colour Analysis

Statistic	Mean R	Mean G	Mean B	Brightness
Mean	194.70	139.26	145.49	159.82
Std Dev	25.06	21.86	24.24	19.18

Statistic	Mean R	Mean G	Mean B	Brightness
Min	63.05	52.03	55.46	63.63
Max	254.29	224.03	238.21	231.77

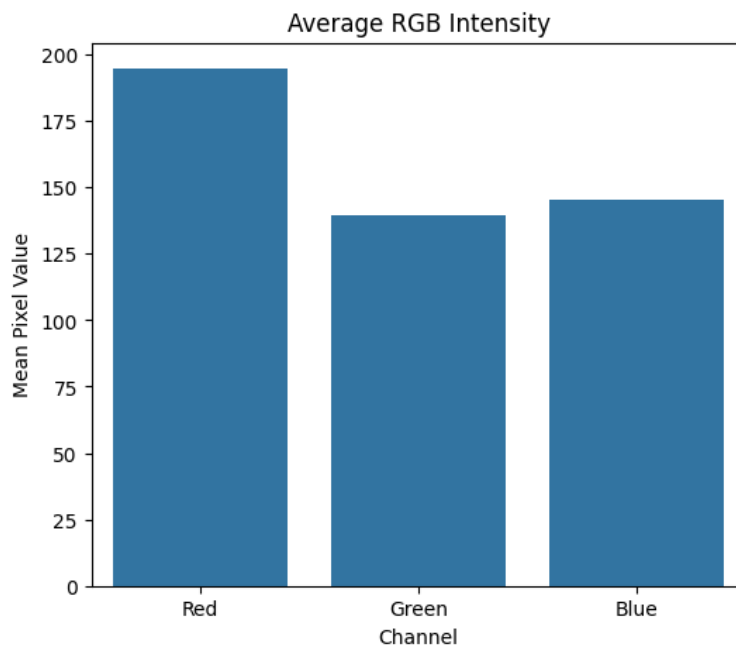


Figure 9. Average RGB intensity across the dataset

Red is the dominant channel on average, consistent with the reddish tones typical of dermoscopic skin imagery, followed by blue and green.

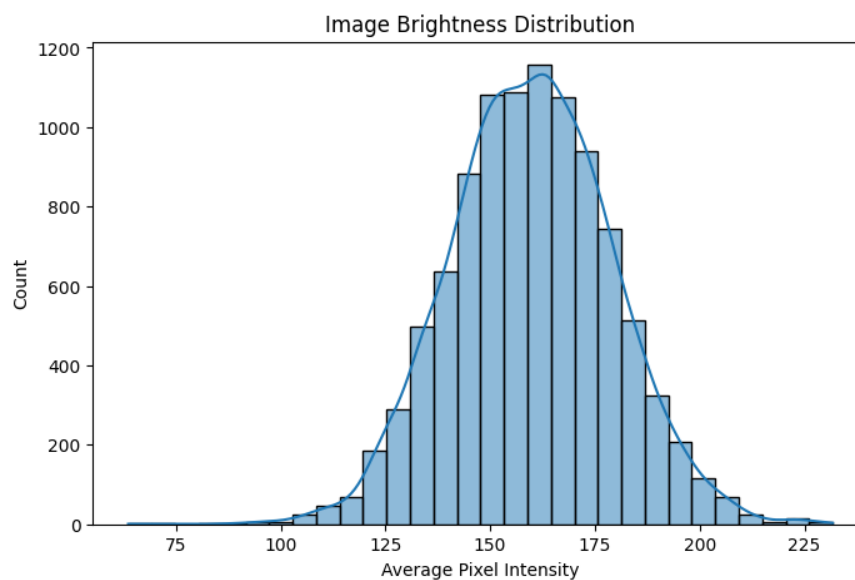


Figure 10. Overall brightness distribution

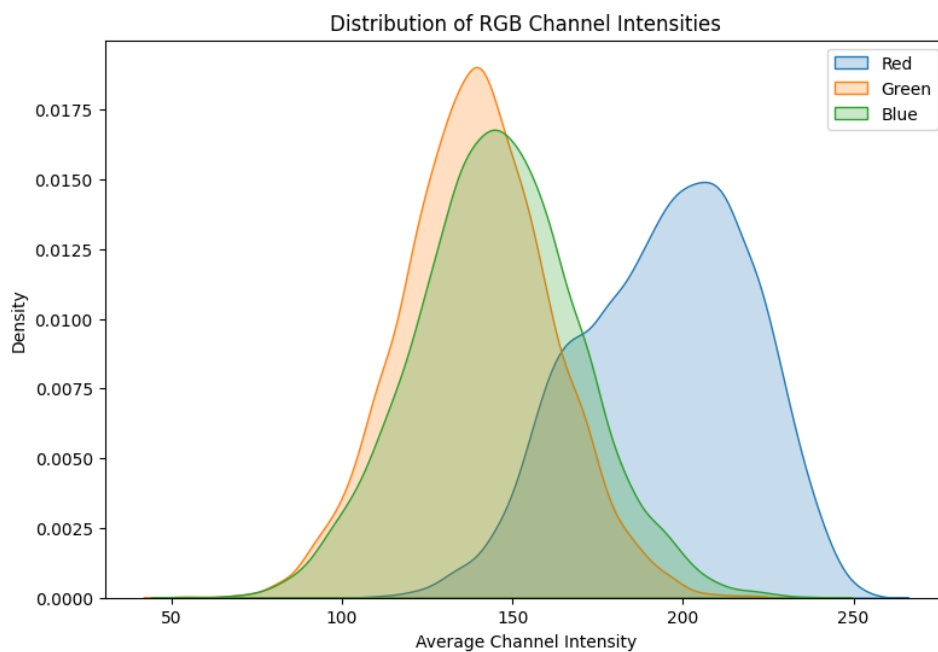


Figure 11. RGB channel intensity distributions (KDE)

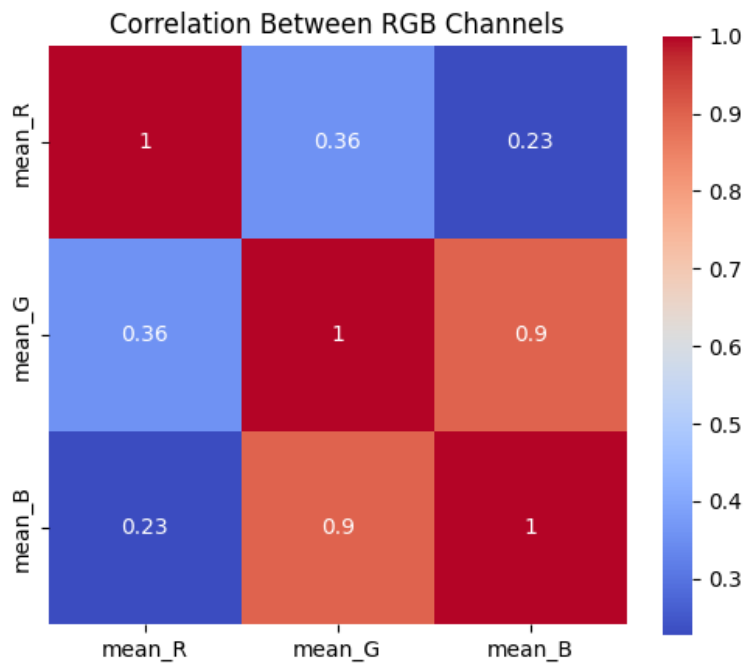


Figure 12. Correlation between RGB channels

The RGB channels are strongly positively correlated with one another, meaning brighter images tend to be brighter across all three channels rather than showing independent per-channel variation.

8.1 Average Colour per Lesion Class

Class	Mean R	Mean G	Mean B
akiec	198.40	149.14	154.72
bcc	195.26	155.76	162.42
bk1	178.19	139.45	147.21
df	198.78	150.93	159.22
mel	177.84	137.95	144.90
nv	199.83	137.30	142.96
vasc	198.40	148.86	162.27

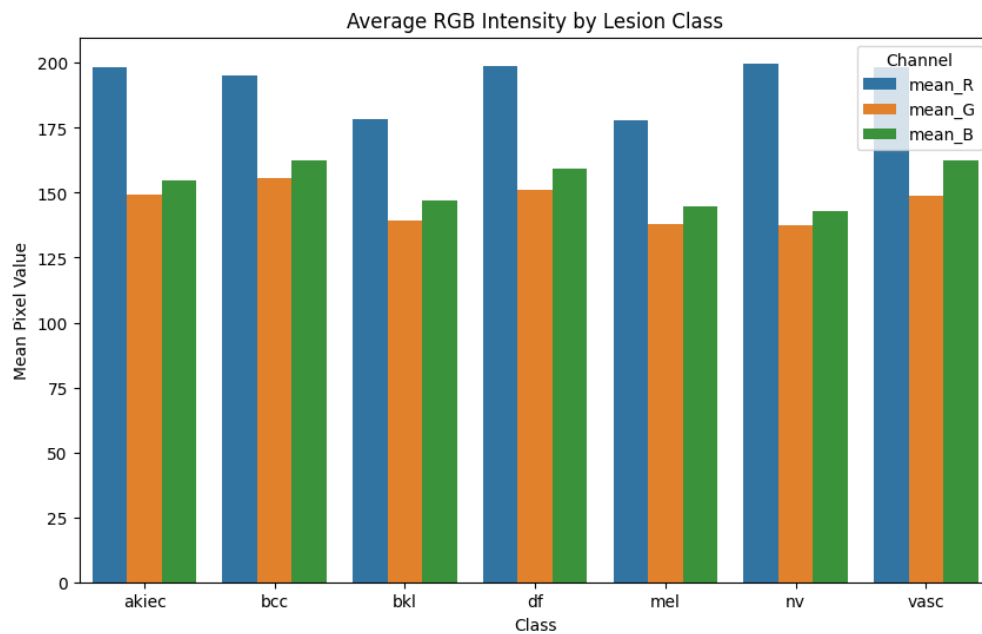


Figure 13. Average RGB intensity broken down by lesion class

9. Image Quality Analysis

9.1 Sharpness (Variance of Laplacian)

Sharpness scores range widely, indicating that most images are moderately sharp but a subset is noticeably blurrier or sharper than the rest.

Statistic	Sharpness (Var. of Laplacian)
count	10,015
mean	79.23
std	124.97

Statistic	Sharpness (Var. of Laplacian)
min	7.62
25%	34.37
50%	49.24
75%	89.46
max	3,825.02

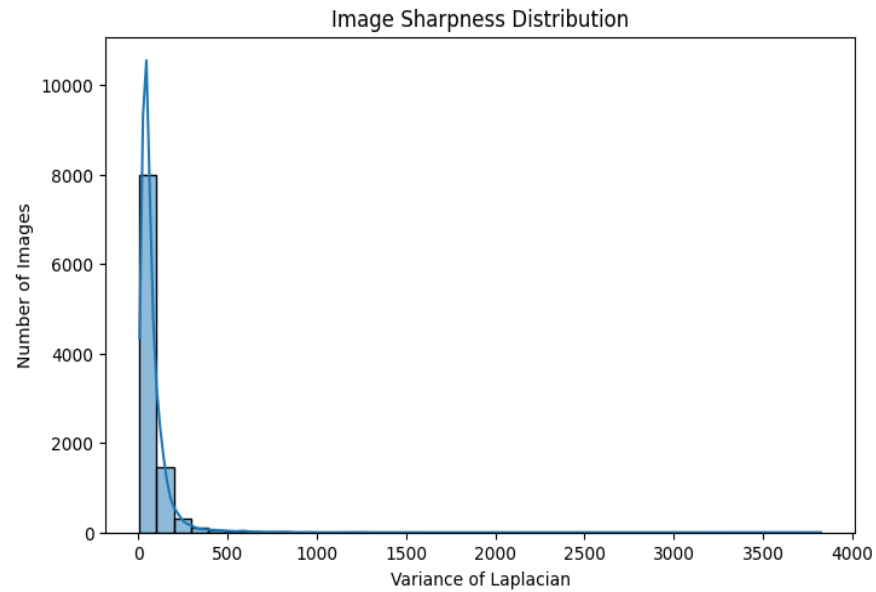


Figure 14. Sharpness distribution across the dataset

9.2 Contrast

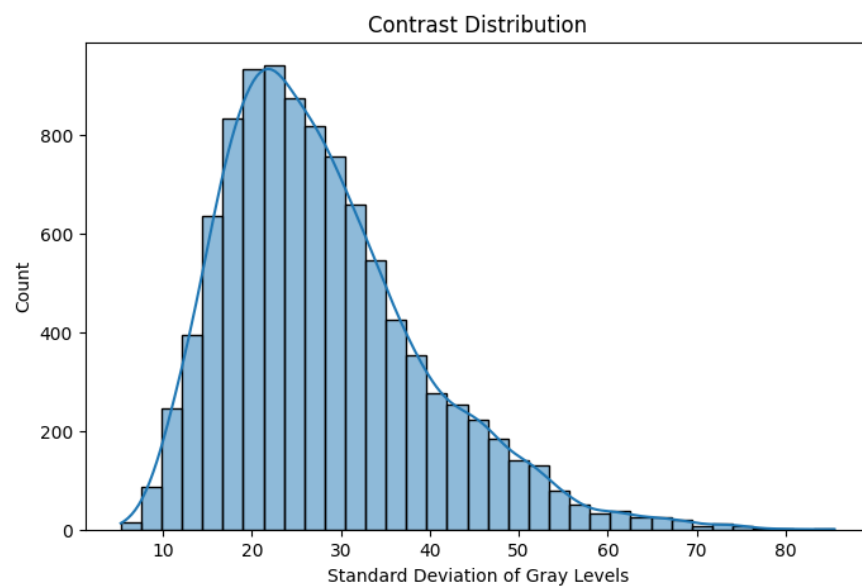


Figure 15. Contrast (grayscale standard deviation) distribution

9.3 Brightness vs. Sharpness

No strong linear relationship is apparent between brightness and sharpness — the two properties vary largely independently across images.

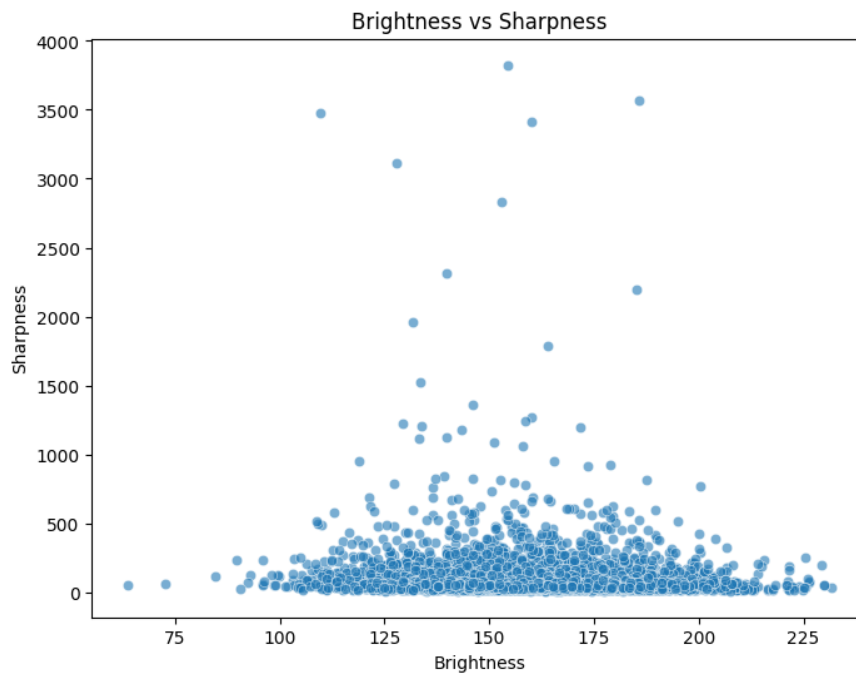


Figure 16. Brightness vs. sharpness scatter plot

9.4 Extreme Examples

Top 5 Sharpest Images



Figure 17. Top 5 sharpest images

Top 5 Blurriest Images

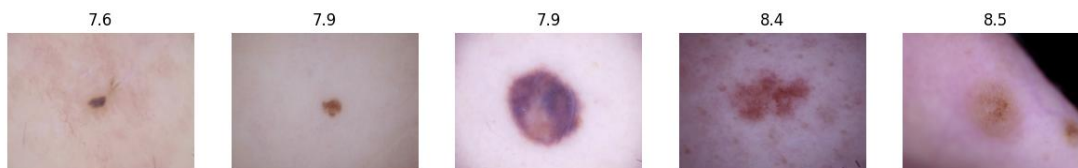


Figure 18. Top 5 blurriest images

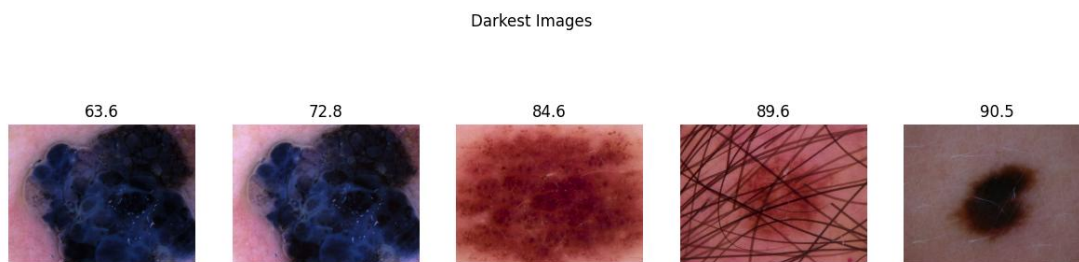


Figure 19. Darkest images in the dataset



Figure 20. Brightest images in the dataset

10. Interactive Visualizations

The notebook also builds a set of interactive Plotly charts covering the same core distributions. Static renders of each are included below.

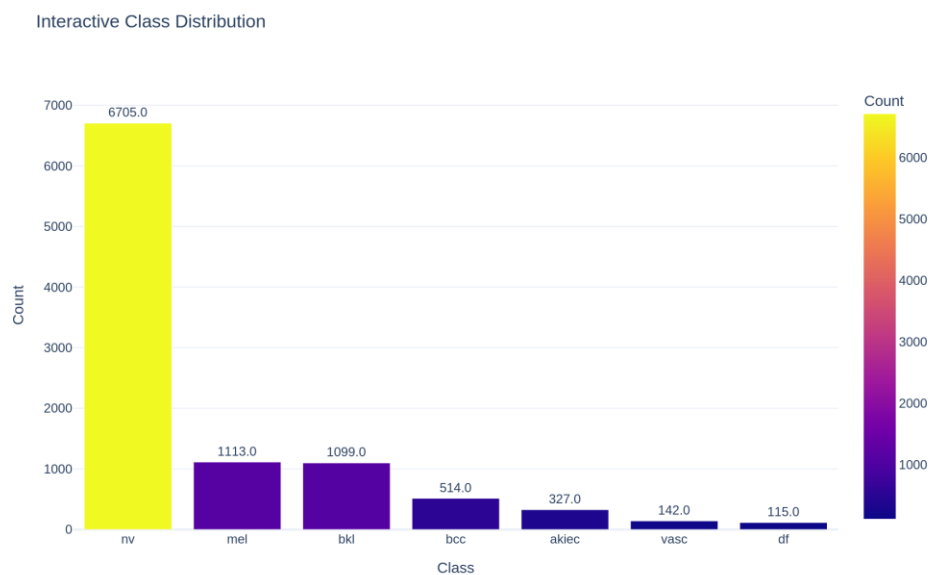


Figure 21. Interactive class distribution (bar)

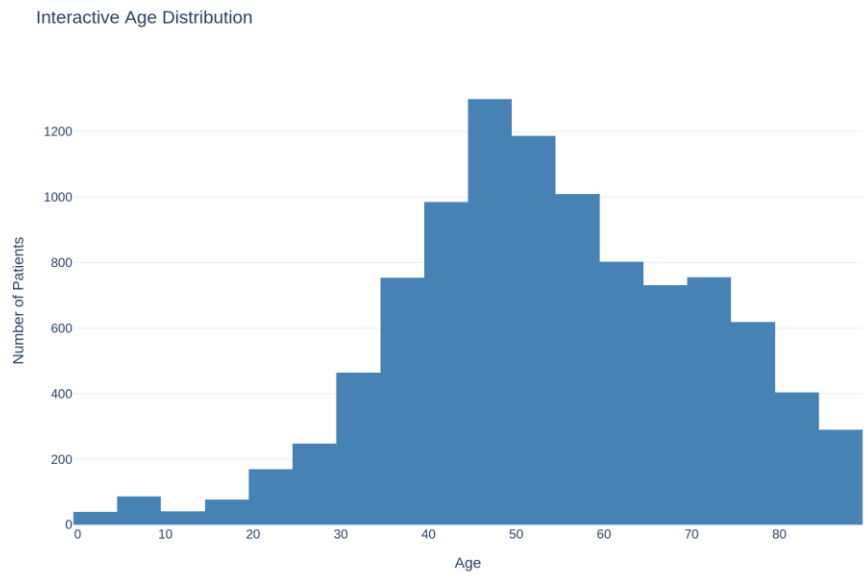


Figure 22. Interactive age distribution (histogram)

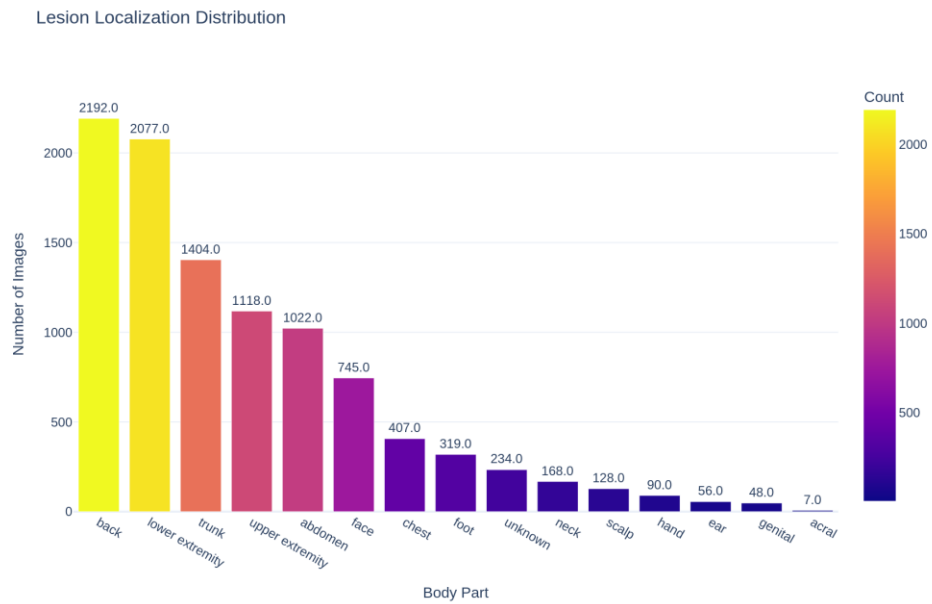


Figure 23. Interactive lesion localization



Figure 24. Interactive brightness vs. sharpness scatter (colored by class)



Figure 25. Interactive RGB channel distribution (box plot)

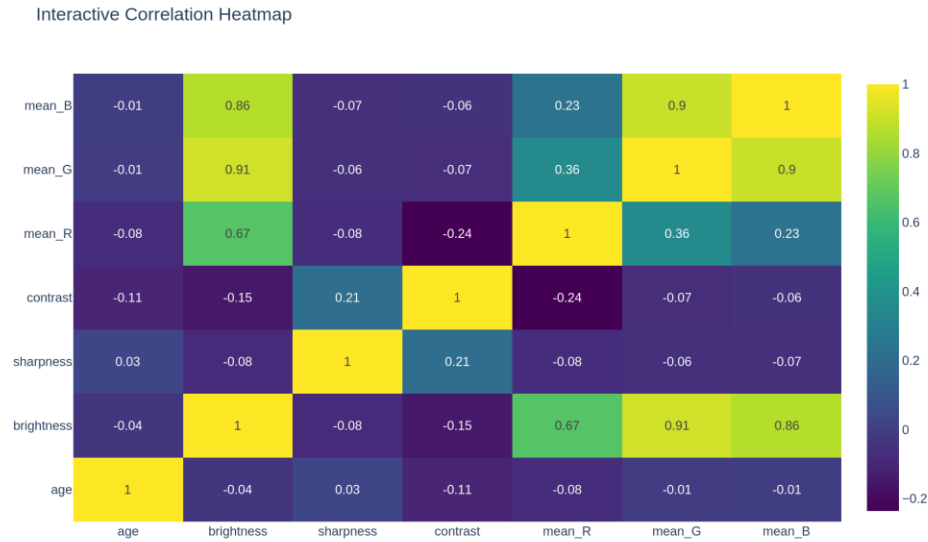


Figure 26. Interactive correlation heatmap (age, brightness, sharpness, contrast, RGB channels)

11. Final EDA Summary

- The HAM10000 dataset contains 10,015 dermoscopic RGB images distributed across seven skin lesion classes.
- The dataset is highly imbalanced, with the Melanocytic Nevus (NV) class accounting for approximately two-thirds of all images, while Dermatofibroma (DF) is the smallest class.
- Most images share a consistent resolution of 600×450 pixels, simplifying preprocessing.
- Several lesion classes exhibit strong visual similarities, particularly Melanoma (MEL) and Nevus (NV), which may increase classification difficulty.
- RGB analysis indicates that red is the dominant colour channel, consistent with dermoscopic skin images.
- Brightness, contrast, and sharpness analyses show that the majority of images are of acceptable quality, although minor illumination and focus variations exist.
- The exploratory analysis suggests that colour, texture, and structural information will all be important for accurate classification.
- The severe class imbalance indicates that imbalance-handling strategies such as weighted loss functions or oversampling should be considered during model development.
- These observations support the use of graph-based representations in later stages, where relationships between image regions can capture structural characteristics beyond conventional pixel-level features.